

Concept Question 10.1 Key numbers for Exoplanet Science

Working with estimations and scaling behaviour is crucial for physicists and astronomers.

- a) Fill in the following lists and remember the scaling ratios between the values, as these will prove crucial when working with exoplanets:

- Distances:

$$1 \text{ AU} = \text{-----} \text{ km}$$

$$1 \text{ AU} = \text{-----} R_{\odot}$$

$$1 R_{\odot} = \text{-----} R_J$$

$$1 R_J = \text{-----} R_{\oplus}$$

- Masses:

$$1 M_{\odot} = \text{-----} \text{ kg}$$

$$1 M_{\odot} = \text{-----} M_J$$

$$1 M_J = \text{-----} M_{\oplus}$$

- Densities:

$$\rho_{\odot}(\text{average}) = \text{-----} \text{ g/cm}^3$$

$$\rho_{\odot}(\text{core}) = \text{-----} \text{ g/cm}^3$$

$$\rho_J(\text{average}) = \text{-----} \text{ g/cm}^3$$

$$\rho_{\oplus}(\text{average}) = \text{-----} \text{ g/cm}^3$$

Hint: you can use the unit converter tool on DACE: <https://dace.unige.ch/unitConverter/>.

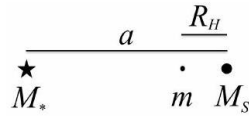
DACE, which stands for Data & Analysis Center for Exoplanets, is a platform for the visualisation, exchange and analysis of extrasolar planet data, hosted by the University of Geneva.

- b) Give the equation for Kepler's third law and solve it for G . Then, use only the numbers in the previous point and your knowledge of the system Sun-Earth to calculate the value of G in units of $\text{AU}^3 M_{\odot}^{-1} \text{yr}^{-2}$.
- c) A Hot Jupiter is orbiting a solar-like star at the same distance as an Earth-twin is orbiting another solar-like star. Estimate the relation between the two periods (no calculations).

Problem 10.2 Hill Radius and limitation of gas accretion

The gas accretion phase, essential for the formation of giant gas planets like Jupiter, is limited by two essential factors which are expressed as the Hill and Bondi radii, R_H and R_B . The radius in which gas can be captured is, depending on the scenario, then given as $R_{gc} = \min(R_H, R_B)$.

- a) What are the Hill and Bondi radii and what do they describe?
- b) A solid body is orbiting a star in a rotating frame (in which the body is resting) in the following configuration:



Assume we put a test mass m into the L_1 Lagrangian point, corresponding to the Hill Radius R_H . Use the balance of forces due to gravitation and rotation in this point, i.e. $\Sigma F = 0$, and derive the expression for R_H in dependency of a , M_* and M_s (the solid body's mass).

Hint: Take care to express the influence of the rotating frame, given by Ω , with the correct dependencies. Use that $R_H \ll a$ at the appropriate steps in your calculation.

- c) Calculate their growth factors $1/\tau_H$ and $1/\tau_B$. How do they depend on the mass of the accreting body?

Hint: The growth timescale can be calculated as

$$\tau = M \cdot \left(\frac{dM}{dt} \right)^{-1} \quad (1)$$

whereby the gas mass accretion rate is given by

$$\frac{dM}{dt} \sim \frac{\Sigma}{H} \Omega R_{gc}^3 \quad (2)$$